

JOINS & SET OPERATIONS

1. Create Table

The screenshot shows the SQL Server Enterprise Manager interface. The left pane displays the 'SCHEMAS' tree with the following structure:

- bookflow_db
- klu
- rahul_2520090016
 - Tables
 - bank_transactions
 - Views
 - Stored Procedures
 - Functions
- sys
- workflow_db

The right pane shows the SQL script for creating the database and tables:

```
1 CREATE DATABASE IF NOT EXISTS Rahul_2520090016;
2 use Rahul_2520090016;
3 DROP TABLE IF EXISTS class_info;
4 DROP TABLE IF EXISTS class;
5 CREATE TABLE class (
6     id INT,
7     name VARCHAR(30)
8 );
9
10 CREATE TABLE class_info (
11     id INT,
12     address VARCHAR(30)
13 );
14
15
16
17
18
19
20
21
22
23
24
25
```

The bottom pane shows the 'Output' window with the following data:

#	Time	Action	Message
7	16:26:50	CREATE DATABASE IF NOT EXISTS Rahul_2520090016	1 row(s) affected, 1 warning(s): 1007 Can't create database h
8	16:26:53	use Rahul_2520090016	0 row(s) affected
9	16:26:58	DROP TABLE IF EXISTS class_info	0 row(s) affected
10	16:27:01	DROP TABLE IF EXISTS class	0 row(s) affected
11	16:27:26	CREATE TABLE class (id INT, name VARCHAR(30))	0 row(s) affected

2: INSERT DATA

Database interface showing SQL execution results for a database named `Rahul_2520090016`.

SCHEMAS

- bookflow_db
- klu
- rahul_2520090016**
 - Tables
 - bank_transactions
 - Views
 - Stored Procedures
 - Functions
 - sys
 - workflow_db

SQL File 5*

```

1 CREATE DATABASE IF NOT EXISTS Rahul_2520090016;
2 use Rahul_2520090016;
3 DROP TABLE IF EXISTS class_info;
4 DROP TABLE IF EXISTS class;
5 CREATE TABLE class (
6     id INT,
7     name VARCHAR(30)
8 );
9
10 CREATE TABLE class_info (
11     id INT,
12     address VARCHAR(30)
13 );
14 INSERT INTO class VALUES
15 (1, 'abhi'),
16 (2, 'adam'),
17 (4, 'alex');
18
19 INSERT INTO class_info VALUES
20 (1, 'DELHI'),
21 (2, 'MUMBAI'),
22 (3, 'CHENNAI');
23
24
25

```

Output

#	Time	Action	Message
10	16:27:01	DROP TABLE IF EXISTS class	0 row(s) affected
11	16:27:26	CREATE TABLE class (id INT, name VARCHAR(30))	0 row(s) affected
12	16:33:48	INSERT INTO class VALUES (1, 'abhi'), (2, 'adam'), (4, 'alex')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0

CROSS JOIN

Database interface showing SQL execution results for a database named `Rahul_2520090016`.

SCHEMAS

- bookflow_db
- klu
- rahul_2520090016**
 - Tables
 - bank_transactions
 - Views
 - Stored Procedures
 - Functions
 - sys
 - workflow_db

SQL File 5*

```

18
19 INSERT INTO class_info VALUES
20 (1, 'DELHI'),
21 (2, 'MUMBAI'),
22 (3, 'CHENNAI');
23 SELECT *FROM class
24 CROSS JOIN class_info;
25

```

Result Grid

id	name
1	abhi
2	adam
4	alex

Output

#	Time	Action	Message
16	16:36:37	CROSS JOIN class_info	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'CROSS JOIN class_info' at line 24
17	16:36:42	SELECT *FROM class LIMIT 0, 1000	3 row(s) returned

PART 2: INNER JOIN

The screenshot shows the SQL Developer interface with a script containing several SQL statements. The output window at the bottom displays the execution results, including error messages for syntax and table existence.

```
20 (1, 'DELHI'),
21 (2, 'MUMBAI'),
22 (3, 'CHENNAI');
23 SELECT *FROM class
24 CROSS JOIN class_info;
25 CREATE TABLE class (
26     id INT,
27     name VARCHAR(30)
28 );
29
30 CREATE TABLE class_info (
31     id INT,
32     address VARCHAR(30)
33 );
34
35 INSERT INTO class VALUES
36 (1, 'abhi'),
37 (2, 'adam'),
38 (3, 'alex'),
39 (4, 'anu');
40
41 INSERT INTO class_info VALUES
42 (1, 'DELHI'),
43 (2, 'MUMBAI'),
44 (3, 'CHENNAI');
```

Output:

#	Time	Action	Message
24	16:45:04	INSERT INTO class_info VALUES	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to
25	16:45:08	SELECT *FROM class LIMIT 0, 1000	3 row(s) returned
26	16:45:43	CREATE TABLE class (id INT, name VARCHAR(30))	Error Code: 1050. Table 'class' already exists
27	16:45:54	CREATE TABLE class_info (id INT, address VARCHAR(30))	0 row(s) affected
28	16:46:02	INSERT INTO class VALUES (1,'abhi'), (2,'adam'), (3,'alex'), (4,'anu')	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0
29	16:46:15	INSERT INTO class_info VALUES (1,'DELHI'), (2,'MUMBAI'), (3,'CHENNAI')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0

INNER JOIN

The screenshot shows the SQL Developer interface with a script containing an INNER JOIN query. The result grid at the bottom displays the output of the query, showing the combination of rows from the 'class' and 'class_info' tables.

```
44 (3, 'CHENNAI');
45 SELECT *FROM class INNER JOIN class_info
46 ON class.id = class_info.id;
47
48
49
50
```

Result Grid:

	id	name	id	address
▶	1	abhi	3	CHENNAI
	1	abhi	2	MUMBAI
	1	abhi	1	DELHI
	2	adam	3	CHENNAI
	2	adam	2	MUMBAI
	2	adam	1	DELHI
	4	alex	3	CHENNAI
	4	alex	2	MUMBAI
	4	alex	1	DELHI
	1	abhi	3	CHENNAI
	1	abhi	2	MUMBAI
	1	abhi	1	DELHI
	2	adam	3	CHENNAI
	2	adam	2	MUMBAI
	2	adam	1	DELHI
	4	alex	3	CHENNAI

INNER JOIN

The screenshot shows a database IDE interface with a menu bar (File, Edit, View, Query, Database, Server, Tools, Scripting, Help) and a toolbar. The left sidebar contains a 'SCHEMAS' tree with a search filter and a list of databases: bookflow_db, klu, rahul_2520090016 (expanded to show Tables, Views, Stored Procedures, Functions), sys, and workflow_db. The main editor displays an SQL query in 'SQL File 6':

```
37
38 • SELECT
39     class.name,
40     class_info.address
41 FROM class
42 INNER JOIN class_info
43 ON class.id = class_info.id;
```

Below the query is the 'Result Grid' showing 3 rows:

	name	address
▶	abhi	DELHI
	adam	MUMBAI
	alex	CHENNAI

The bottom section shows the 'Output' pane with 'Action Output' selected, displaying a log of SQL actions and their results:

#	Time	Action	Message
64	18:25:56	INNER JOIN class_info	Error Code: 1064. You have an error in your
65	18:26:00	ON class.id = class_info.id	Error Code: 1064. You have an error in your
66	18:26:13	SELECT class.name, class_info.address FROM class INNER JOIN class_info ON class.id = class_info.id ...	3 row(s) returned
67	18:26:23	SELECT *FROM class INNER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	3 row(s) returned
68	18:26:33	SELECT * FROM class NATURAL JOIN class_info LIMIT 0, 1000	3 row(s) returned
69	18:26:49	SELECT class.name, class_info.address FROM class INNER JOIN class_info ON class.id = class_info.id ...	3 row(s) returned

PART 3: NATURAL JOIN

SQL Editor Interface showing a query execution error.

Navigator: Schemas, Filter objects, bookflow_db, klu, rahul_2520090016, Tables, bank_transactions, Views, Stored Procedures, Functions, sys, workflow_db.

SQL Editor: skill all cammonds*, bank_transactions, SQL File 4*, SQL File 5*, SQL File 6*.

Query:

```

43 ON class.id = class_info.id;
44
45
46 SELECT *
47 FROM class
48 NATURAL JOIN class_info;
49

```

Result Grid:

id	name	address
1	abhi	DELHI
2	adam	MUMBAI
3	alex	CHENNAI

Output:

#	Time	Action	Message
65	18:26:00	ON class.id = class_info.id	Error Code: 1064. You have an error in your SQL syntax; check the manual that come
66	18:26:13	SELECT class.name, class_info.address FROM class INNER JOIN class_info ON class.id = class_info.id ...	3 row(s) returned
67	18:26:23	SELECT * FROM class INNER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	3 row(s) returned
68	18:26:33	SELECT * FROM class NATURAL JOIN class_info LIMIT 0, 1000	3 row(s) returned
69	18:26:49	SELECT class.name, class_info.address FROM class INNER JOIN class_info ON class.id = class_info.id ...	3 row(s) returned
70	18:29:09	SELECT * FROM class NATURAL JOIN class_info LIMIT 0, 1000	3 row(s) returned

PART 4: LEFT OUTER JOIN

SQL Editor Interface showing a successful query execution.

Navigator: Schemas, Filter objects, bookflow_db, klu, rahul_2520090016, Tables, bank_transactions, Views, Stored Procedures, Functions, sys, workflow_db.

SQL Editor: skill all cammonds*, bank_transactions, SQL File 4*, SQL File 5*, SQL File 6*.

Query:

```

44
45
46 SELECT *
47 FROM class
48 NATURAL JOIN class_info;
49
50
51 INSERT INTO class VALUES
52 (5, 'ashish');
53
54 INSERT INTO class_info VALUES
55 (7, 'NOIDA'),
56 (8, 'PANIPAT');
57
58
59
60
61
62
63
64
65
66

```

Output:

#	Time	Action	Message
67	18:26:23	SELECT * FROM class INNER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	3 row(s) returned
68	18:26:33	SELECT * FROM class NATURAL JOIN class_info LIMIT 0, 1000	3 row(s) returned
69	18:26:49	SELECT class.name, class_info.address FROM class INNER JOIN class_info ON class.id = class_info.id ...	3 row(s) returned
70	18:29:09	SELECT * FROM class NATURAL JOIN class_info LIMIT 0, 1000	3 row(s) returned
71	18:30:52	INSERT INTO class VALUES (5, 'ashish')	1 row(s) affected
72	18:30:58	INSERT INTO class_info VALUES (7, 'NOIDA'), (8, 'PANIPAT')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0

LEFT JOIN

The screenshot shows the SQL Developer interface with a query window containing the following SQL code:

```
52 (5, 'ashish');
53
54 INSERT INTO class_info VALUES
55 (7, 'NOIDA'),
56 (8, 'PANIPAT');
57
58
```

The Results window displays the following data:

id	name	id	address
1	abhi	1	DELHI
2	adam	2	MUMBAI
3	alex	3	CHENNAI
4	anu	NULL	NULL
5	ashish	NULL	NULL

The Output window shows the following messages:

#	Time	Action	Message
68	18:26:33	SELECT * FROM class NATURAL JOIN class_info LIMIT 0, 1000	3 row(s) returned
69	18:26:49	SELECT class.name, class_info.address FROM class INNER JOIN class_info ON class.id = class_info.id ...	3 row(s) returned
70	18:29:09	SELECT * FROM class NATURAL JOIN class_info LIMIT 0, 1000	3 row(s) returned
71	18:30:52	INSERT INTO class VALUES (5, 'ashish')	1 row(s) affected
72	18:30:58	INSERT INTO class_info VALUES (7, 'NOIDA'), (8, 'PANIPAT')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
73	18:32:33	SELECT * FROM class LEFT OUTER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	5 row(s) returned

LEFT JOIN

The screenshot shows the SQL Developer interface with a query window containing the following SQL code:

```
60 SELECT *
61 FROM class
62 LEFT JOIN class_info
63 ON class.id = class_info.id
64 WHERE class_info.id IS NULL;
65
66
```

The Results window displays the following data:

id	name	id	address
4	anu	NULL	NULL
5	ashish	NULL	NULL

The Output window shows the following messages:

#	Time	Action	Message
69	18:26:49	SELECT class.name, class_info.address FROM class INNER JOIN class_info ON class.id = class_info.id ...	3 row(s) returned
70	18:29:09	SELECT * FROM class NATURAL JOIN class_info LIMIT 0, 1000	3 row(s) returned
71	18:30:52	INSERT INTO class VALUES (5, 'ashish')	1 row(s) affected
72	18:30:58	INSERT INTO class_info VALUES (7, 'NOIDA'), (8, 'PANIPAT')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
73	18:32:33	SELECT * FROM class LEFT OUTER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	5 row(s) returned
74	18:34:09	SELECT * FROM class LEFT JOIN class_info ON class.id = class_info.id WHERE class_info.id IS NULL LIMIT ...	2 row(s) returned

PART 5: RIGHT OUTER JOIN

RIGHT JOIN

The screenshot displays a database management interface with a SQL query editor and a results grid. The query is as follows:

```
81 • SELECT *
82 FROM class
83 RIGHT OUTER JOIN class_info
84 ON class.id = class_info.id;
85
86
87
```

The results grid shows the following data:

	id	name	id	address
▶	1	abhi	1	DELHI
	2	adam	2	MUMBAI
	3	alex	3	CHENNAI
	NULL	NULL	7	NOIDA
	NULL	NULL	8	PANIPAT

The bottom section of the interface shows the 'Action Output' log with the following entries:

#	Time	Action	Message
70	18:29:09	SELECT * FROM class NATURAL JOIN class_info LIMIT 0, 1000	3 row(s) returned
71	18:30:52	INSERT INTO class VALUES (5, 'ashish')	1 row(s) affected
72	18:30:58	INSERT INTO class_info VALUES (7, (5, 'ashish'))	2 row(s) affected Records: 2 Duplicates:
73	18:32:33	SELECT * FROM class LEFT OUTER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	5 row(s) returned
74	18:34:09	SELECT * FROM class LEFT JOIN class_info ON class.id = class_info.id WHERE class_info.id IS NULL LIMIT...	2 row(s) returned
75	18:51:49	SELECT * FROM class RIGHT OUTER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	5 row(s) returned

SQL File 6* x

skill all cammonds* bank_transactions SQL File 4* SQL File 5* SQL File 6* x

Limit to 1000 rows

```

97
98
99 • SELECT *
100 FROM class
101 RIGHT JOIN class_info
102 ON class.id = class_info.id
103 WHERE class.id IS NULL;

```

Result Grid

	id	name	id	address
	NULL	NULL	7	NOIDA
	NULL	NULL	8	PANIPAT

Administration Schemas

Information

No object selected

Result 11 x

Output

Action Output

#	Time	Action	Message
71	18:30:52	INSERT INTO class VALUES (5, 'ashish')	1 row(s) affected
72	18:30:58	INSERT INTO class_info VALUES (7, 'NOIDA'), (8, 'PANIPAT')	2 row(s) affected Records: 2 Du
73	18:32:33	SELECT * FROM class LEFT OUTER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	5 row(s) returned
74	18:34:09	SELECT * FROM class LEFT JOIN class_info ON class.id = class_info.id WHERE class_info.id IS NULL LIMIT...	2 row(s) returned
75	18:51:49	SELECT * FROM class RIGHT OUTER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	5 row(s) returned
76	18:53:07	SELECT * FROM class RIGHT JOIN class_info ON class.id = class_info.id WHERE class.id IS NULL LIMIT 0, ...	2 row(s) returned

Object Info Session

PART 6: FULL OUTER JOIN

SQL File 6

```

113
114 • SELECT
115     class.id AS class_id,
116     class.name,
117     class_info.id AS info_id,
118     class_info.address
119 FROM class

```

Result Grid

class_id	name	info_id	address
1	abhi	1	DELHI
2	adam	2	MUMBAI
3	alex	3	CHENNAI
4	anu	NULL	NULL
5	ashish	NULL	NULL
7		7	NOIDA
8		8	PANIPAT

Output

#	Time	Action	Message
72	18:30:58	INSERT INTO class_info VALUES (7, 'NOIDA'), (8, 'PANIPAT')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
73	18:32:33	SELECT * FROM class LEFT OUTER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	5 row(s) returned
74	18:34:09	SELECT * FROM class LEFT JOIN class_info ON class.id = class_info.id WHERE class_info.id IS NULL LIMIT...	2 row(s) returned
75	18:51:49	SELECT * FROM class RIGHT OUTER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	5 row(s) returned
76	18:53:07	SELECT * FROM class RIGHT JOIN class_info ON class.id = class_info.id WHERE class.id IS NULL LIMIT 0, ...	2 row(s) returned
77	18:54:14	SELECT class.id AS class_id, class.name, class_info.id AS info_id, class_info.address FROM class L...	7 row(s) returned

FULL OUTER JOIN

SQL File 6

```

134 • SELECT
135     class.id AS class_id,
136     class.name,
137     class_info.id AS info_id,
138     class_info.address
139 FROM class
140 LEFT JOIN class_info

```

Result Grid

class_id	name	info_id	address
4	anu	NULL	NULL
5	ashish	NULL	NULL
7		7	NOIDA
8		8	PANIPAT

Output

#	Time	Action	Message
73	18:32:33	SELECT * FROM class LEFT OUTER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	5 row(s) returned
74	18:34:09	SELECT * FROM class LEFT JOIN class_info ON class.id = class_info.id WHERE class_info.id IS NULL LIMIT...	2 row(s) returned
75	18:51:49	SELECT * FROM class RIGHT OUTER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	5 row(s) returned
76	18:53:07	SELECT * FROM class RIGHT JOIN class_info ON class.id = class_info.id WHERE class.id IS NULL LIMIT 0, ...	2 row(s) returned
77	18:54:14	SELECT class.id AS class_id, class.name, class_info.id AS info_id, class_info.address FROM class L...	7 row(s) returned
78	18:56:05	SELECT class.id AS class_id, class.name, class_info.id AS info_id, class_info.address FROM class L...	4 row(s) returned

PART 7: UNION

The screenshot shows the SQL Enterprise Manager interface with the following SQL script in the editor:

```
153 ON class.id = class_info.id
154 WHERE class.id IS NULL;
155
156
157 DROP TABLE IF EXISTS first_table;
158 DROP TABLE IF EXISTS second_table;
159
160 CREATE TABLE first_table (
161     id INT,
162     name VARCHAR(30)
163 );
164
165 CREATE TABLE second_table (
166     id INT,
167     name VARCHAR(30)
168 );
169
170 INSERT INTO first_table VALUES
171 (1, 'abhi'),
172 (2, 'adam');
173
174 INSERT INTO second_table VALUES
175 (2, 'adam'),
176 (3, 'chester');
```

The Output pane shows the following results:

#	Time	Action	Message
79	18:57:19	DROP TABLE IF EXISTS first_table	0 row(s) affected, 1 warning(s): 1051 Unknown table 'joins_set_operations.first_table'
80	18:57:19	DROP TABLE IF EXISTS second_table	0 row(s) affected, 1 warning(s): 1051 Unknown table 'joins_set_operations.second_table'
81	18:57:19	CREATE TABLE first_table (id INT, name VARCHAR(30))	0 row(s) affected
82	18:57:19	CREATE TABLE second_table (id INT, name VARCHAR(30))	0 row(s) affected
83	18:57:19	INSERT INTO first_table VALUES (1, 'abhi'), (2, 'adam')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
84	18:57:19	INSERT INTO second_table VALUES (2, 'adam'), (3, 'chester')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0

The screenshot shows the SQL Enterprise Manager interface with the following SQL script in the editor:

```
179 SELECT *
180 FROM first_table
181 UNION
182 SELECT *
183 FROM second_table;
184
185
```

The Result Grid shows the following data:

id	name
1	abhi
2	adam
3	chester

The Output pane shows the following results:

#	Time	Action	Message
80	18:57:19	DROP TABLE IF EXISTS second_table	0 row(s) affected, 1 warning(s): 1051 Unknown table 'joins_se
81	18:57:19	CREATE TABLE first_table (id INT, name VARCHAR(30))	0 row(s) affected
82	18:57:19	CREATE TABLE second_table (id INT, name VARCHAR(30))	0 row(s) affected
83	18:57:19	INSERT INTO first_table VALUES (1, 'abhi'), (2, 'adam')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
84	18:57:19	INSERT INTO second_table VALUES (2, 'adam'), (3, 'chester')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
85	18:58:12	SELECT * FROM first_table UNION SELECT * FROM second_table	3 row(s) returned

SQL File 6*

```

192
193
194 • SELECT name
195   FROM first_table
196  UNION
197  SELECT name
198   FROM second_table;

```

Result Grid

name
abhi
adam
chester

Output

Action Output

#	Time	Action	Message
81	18:57:19	CREATE TABLE first_table (id INT, name VARCHAR(30))	0 row(s) affected
82	18:57:19	CREATE TABLE second_table (id INT, name VARCHAR(30))	0 row(s) affected
83	18:57:19	INSERT INTO first_table VALUES (1, 'abhi'), (2, 'adam')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
84	18:57:19	INSERT INTO second_table VALUES (2, 'adam'), (3, 'chester')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
85	18:58:12	SELECT * FROM first_table UNION SELECT * FROM second_table	3 row(s) returned
86	18:59:15	SELECT name FROM first_table UNION SELECT name FROM second_table	3 row(s) returned

PART 8: UNION ALL

SQL File 6*

```

198   FROM second_table;
199
200 • SELECT *
201   FROM first_table
202  UNION ALL
203  SELECT *
204   FROM second_table;

```

Result Grid

id	name
1	abhi
2	adam
2	adam
3	chester

Output

Action Output

#	Time	Action	Message
82	18:57:19	CREATE TABLE second_table (id INT, name VARCHAR(30))	0 row(s) affected
83	18:57:19	INSERT INTO first_table VALUES (1, 'abhi'), (2, 'adam')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
84	18:57:19	INSERT INTO second_table VALUES (2, 'adam'), (3, 'chester')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
85	18:58:12	SELECT * FROM first_table UNION SELECT * FROM second_table	3 row(s) returned
86	18:59:15	SELECT name FROM first_table UNION SELECT name FROM second_table	3 row(s) returned
87	19:00:29	SELECT * FROM first_table UNION ALL SELECT * FROM second_table	4 row(s) returned

SQL File 6* x

File Edit View Query Database Server Tools Scripting Help

Navigator skill all cammonds* bank_transactions SQL File 4* SQL File 5* SQL File 6* x

Limit to 1000 rows

```

206
207 • SELECT COUNT(*) AS total_records
208 FROM
209 (
210     SELECT *
211     FROM first_table
212 )

```

Result Grid

total_records
4

Administration Schemas

Information

No object selected

Result 17 x Read Only

Output

Action Output

#	Time	Action	Message
83	18:57:19	INSERT INTO first_table VALUES (1, 'abhi'), (2, 'adam')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
84	18:57:19	INSERT INTO second_table VALUES (2, 'adam'), (3, 'chester')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
85	18:58:12	SELECT * FROM first_table UNION SELECT * FROM second_table	3 row(s) returned
86	18:59:15	SELECT name FROM first_table UNION SELECT name FROM second_table	3 row(s) returned
87	19:00:29	SELECT * FROM first_table UNION ALL SELECT * FROM second_table	4 row(s) returned
88	19:01:23	SELECT COUNT(*) AS total_records FROM (SELECT * FROM first_table UNION ALL SELECT * FROM second_table)	1 row(s) returned

Object Info Session

PART 9: INTERSECT

SQL File 6* x

File Edit View Query Database Server Tools Scripting Help

Navigator skill all cammonds* bank_transactions SQL File 4* SQL File 5* SQL File 6* x

Limit to 1000 rows

```

222
223 • SELECT f.id, f.name
224 FROM first_table f
225 INNER JOIN second_table s
226 ON f.id = s.id
227 AND f.name = s.name;
228

```

Result Grid

id	name
2	adam

Administration Schemas

Information

No object selected

Result 18 x Read Only

Output

Action Output

#	Time	Action	Message
84	18:57:19	INSERT INTO second_table VALUES (2, 'adam'), (3, 'chester')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0
85	18:58:12	SELECT * FROM first_table UNION SELECT * FROM second_table	3 row(s) returned
86	18:59:15	SELECT name FROM first_table UNION SELECT name FROM second_table	3 row(s) returned
87	19:00:29	SELECT * FROM first_table UNION ALL SELECT * FROM second_table	4 row(s) returned
88	19:01:23	SELECT COUNT(*) AS total_records FROM (SELECT * FROM first_table UNION ALL SELECT * FROM second_table)	1 row(s) returned
89	19:02:14	SELECT f.id, f.name FROM first_table f INNER JOIN second_table s ON f.id = s.id AND f.name = s.name LIMIT 1	1 row(s) returned

Object Info Session

SQL File 6

```

240
241
242 • SELECT DISTINCT f.name
243 FROM first_table f
244 INNER JOIN second_table s
245 ON f.name = s.name;
246

```

Result Grid

name
adam

Result 19

Output

Action Output

#	Time	Action	Message
85	18:58:12	SELECT * FROM first_table UNION SELECT * FROM second_table	3 row(s) returned
86	18:59:15	SELECT name FROM first_table UNION SELECT name FROM second_table	3 row(s) returned
87	19:00:29	SELECT * FROM first_table UNION ALL SELECT * FROM second_table	4 row(s) returned
88	19:01:23	SELECT COUNT(*) AS total_records FROM (SELECT * FROM first_table UNION ALL SELECT * ...	1 row(s) returned
89	19:02:14	SELECT f.id, f.name FROM first_table f INNER JOIN second_table s ON f.id = s.id AND f.name = s.name LIM...	1 row(s) returned
90	19:02:51	SELECT DISTINCT f.name FROM first_table f INNER JOIN second_table s ON f.name = s.name LIMIT 0, 1000	1 row(s) returned

PART 10: MINUS

SQL File 6

```

246
247
248 • SELECT f.id, f.name
249 FROM first_table f
250 WHERE NOT EXISTS
251 (
252 SELECT 1

```

Result Grid

id	name
1	abhi
2	adam

first_table 20

Output

Action Output

#	Time	Action	Message
87	19:00:29	SELECT * FROM first_table UNION ALL SELECT * FROM second_table	4 row(s) returned
88	19:01:23	SELECT COUNT(*) AS total_records FROM (SELECT * FROM first_table UNION ALL SELECT * ...	1 row(s) returned
89	19:02:14	SELECT f.id, f.name FROM first_table f INNER JOIN second_table s ON f.id = s.id AND f.name = s.name LIM...	1 row(s) returned
90	19:02:51	SELECT DISTINCT f.name FROM first_table f INNER JOIN second_table s ON f.name = s.name LIMIT 0, 1000	1 row(s) returned
91	19:04:23	SELECT f.id, f.name FROM first_table f WHERE NOT EXISTS	Error Code: 1064. You have an error in your SQL syntax; check the man
92	19:04:33	SELECT f.id, f.name FROM first_table f LIMIT 0, 1000	2 row(s) returned

Rahul_2520090016 x

File Edit View Query Database Server Tools Scripting Help

Navigator: skill all cammonds* bank_transactions SQL File 4* SQL File 5* SQL File 6* x SQL

SCHEMAS

Filter objects

- bookflow_db
- klu
- rahul_2520090016
 - Tables
 - bank_transactions
 - Views
 - Stored Procedures
 - Functions
 - sys
 - workflow_db

284 ELSE 'Address Available'

285 END AS Status

286 FROM class c

287 LEFT JOIN class_info ci

288 ON c.id = ci.idj

289

290 -- DISPLAY TABLES

Result Grid | Filter Rows: | Export: | Wrap Cell Contents: |

	id	name	Status
1	abhi	Address Available	
2	adam	Address Available	
3	alex	Address Available	
4	anu	Address Missing	
5	ashish	Address Missing	

Administration Schemas

Information

No object selected

Result 24 x

Output

Action Output

#	Time	Action	Message
93	19:05:06	WHERE NOT EXISTS (SELECT 1 FROM second_table s WHERE s.id = f.id AND s.name = f.name)	Error Code: 1064. You have an error in your SQL syntax; check the manual t
94	19:05:36	WHERE NOT EXISTS	Error Code: 1064. You have an error in your SQL syntax; check the manual t
95	19:05:41	SELECT f.id, f.name FROM first_table f WHERE NOT EXISTS (SELECT 1 FROM second_table s WH...	1 row(s) returned
96	19:06:02	SELECT DISTINCT f.name FROM first_table f WHERE NOT EXISTS (SELECT 1 FROM second_table s...	1 row(s) returned
97	19:06:13	SELECT c.id, c.name, ci.address FROM class c INNER JOIN class_info ci ON c.id = ci.id LIMIT 0, 1000	3 row(s) returned
98	19:06:27	SELECT c.id, c.name, CASE WHEN ci.address IS NULL THEN 'Address Missing' ELSE...	5 row(s) returned

Object Info Session

VIEWS

MySQL Workbench

Rahul_2520090016 x

File Edit View Query Database Server Tools Scripting Help

Navigator: skill week - 3.4* x

Limit to 1000 rows

SCHEMAS

Filter objects

- 2520090016_rahul
 - bankdb
 - Tables
 - Views
 - Stored Procedures
 - Functions
 - bookflow_db
 - joins_set_operations
 - klu
 - rahul_2520090016
 - sys
 - workflow_db

Administration Schemas

Information

Schema: bookflow_db

```
213
214 • USE BankDB;
215
216 • CREATE VIEW Customer_Account_View AS
217 SELECT
218     C.Customer_ID,
219     C.Customer_Name,
220     C.Phone,
221     C.Email,
222     C.City,
223     A.Account_No,
224     A.Account_Type,
225     A.Balance,
226     A.Branch
227 FROM Customer C
```

Result Grid

	Customer_ID	Customer_Name	Phone	Email	City	Account_No	Account_Type	Balance	Branch
--	-------------	---------------	-------	-------	------	------------	--------------	---------	--------

ccount_View 1 x

Output

Action Output

#	Time	Action	Message
1	20:17:55	USE BankDB	0 row(s) affected
2	20:17:55	CREATE VIEW Customer_Account_View AS SELECT C.Customer_ID, C.Customer_Name, C.Phone, ...	0 row(s) affected
3	20:17:55	SELECT * FROM Customer_Account_View LIMIT 0, 1000	0 row(s) returned

Object Info Session

MySQL Workbench

Rahul_2520090016 x

File Edit View Query Database Server Tools Scripting Help

Navigator: skill week - 3.4 x

SCHEMAS

Filter objects

- 2520090016_rahul
 - bankdb
 - Tables
 - Views
 - Stored Procedures
 - Functions
 - bookflow_db
 - joins_set_operations
 - klu
 - rahul_2520090016
 - sys
 - workflow_db

```
233 CREATE VIEW Customer_Loan_View AS
234 SELECT
235     C.Customer_ID,
236     C.Customer_Name,
237     L.Loan_ID,
238     L.Loan_Type,
239     L.Loan_Amount,
240     L.Interest_Rate
241 FROM Customer C
242 JOIN Loan L
243 ON C.Customer_ID = L.Customer_ID;
244
245 SELECT * FROM Customer_Loan_View;
246
```

Administration Schemas

Information

Schema: bookflow_db

Result Grid

	Customer_ID	Customer_Name	Loan_ID	Loan_Type	Loan_Amount	Interest_Rate
--	-------------	---------------	---------	-----------	-------------	---------------

Customer_Loan_View 2 x

Output

Action Output

#	Time	Action	Message
1	20:17:55	USE BankDB	0 row(s) affected
2	20:17:55	CREATE VIEW Customer_Account_View AS SELECT C.Customer_ID, C.Customer_Name, C.Phone, ...	0 row(s) affected
3	20:17:55	SELECT * FROM Customer_Account_View LIMIT 0, 1000	0 row(s) returned
4	20:18:36	CREATE VIEW Customer_Loan_View AS SELECT C.Customer_ID, C.Customer_Name, L.Loan_ID, L...	0 row(s) affected
5	20:18:36	SELECT * FROM Customer_Loan_View LIMIT 0, 1000	0 row(s) returned

Object Info Session

MySQL Workbench

Rahul_2520090016 x

File Edit View Query Database Server Tools Scripting Help

Navigator skill week - 3.4 x

Limit to 1000 rows

247 CREATE VIEW Transaction_View AS
 248 SELECT
 249 T.Transaction_ID,
 250 T.Account_No,
 251 C.Customer_Name,
 252 T.Transaction_Type,
 253 T.Amount,
 254 T.Transaction_Date
 255 FROM Bank_Transaction T
 256 JOIN Account A
 257 ON T.Account_No = A.Account_No
 258 JOIN Customer C
 259 ON A.Customer_ID = C.Customer_ID;
 260
 261 SELECT * FROM Transaction_View

Administration Schemas

Information

Schema: bookflow_db

Result Grid

Transaction_ID	Account_No	Customer_Name	Transaction_Type	Amount	Transaction_Date
----------------	------------	---------------	------------------	--------	------------------

Transaction_View 3 x

Output

Action Output

#	Time	Action	Message
2	20:17:55	CREATE VIEW Customer_Account_View AS SELECT C.Customer_ID, C.Customer_Name, C.Phone, ...	0 row(s) affected
3	20:17:55	SELECT * FROM Customer_Account_View LIMIT 0, 1000	0 row(s) returned
4	20:18:36	CREATE VIEW Customer_Loan_View AS SELECT C.Customer_ID, C.Customer_Name, L.Loan_ID, ...	0 row(s) affected
5	20:18:36	SELECT * FROM Customer_Loan_View LIMIT 0, 1000	0 row(s) returned
6	20:19:14	CREATE VIEW Transaction_View AS SELECT T.Transaction_ID, T.Account_No, C.Customer_Name, ...	0 row(s) affected
7	20:19:14	SELECT * FROM Transaction_View LIMIT 0, 1000	0 row(s) returned

Object Info Session

TRIGGERS

Trigger 1 — Prevent Negative Balance

MySQL Workbench

Rahul_2520090016

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- 2520090016_rahul
 - bankdb
 - Tables
 - Views
 - Stored Procedures
 - Functions
 - bookflow_db
 - joins_set_operations
 - klu
 - rahul_2520090016
 - sys
 - workflow_db

Administration Schemas

Information

Schema: bookflow_db

```

257 ON T.Account_No = A.Account_No
258 JOIN Customer C
259 ON A.Customer_ID = C.Customer_ID;
260
261 • SELECT * FROM Transaction_View;
262
263 DELIMITER //
264
265 • CREATE TRIGGER CheckBalance
266 BEFORE UPDATE ON Account
267 FOR EACH ROW
268 BEGIN
269     IF NEW.Balance < 0 THEN
270         SIGNAL SQLSTATE '45000'
271         SET MESSAGE_TEXT = 'Transaction failed: Insufficient balance';
272     END IF;
273 END //
274
275 DELIMITER ;
276

```

Output

Action Output

#	Time	Action	Message
1	20:17:55	USE BankDB	0 row(s) affected
2	20:17:55	CREATE VIEW Customer_Account_View AS SELECT C.Customer_ID, C.Customer_Name, C.Phone, ...	0 row(s) affected
3	20:17:55	SELECT * FROM Customer_Account_View LIMIT 0, 1000	0 row(s) returned
4	20:18:36	CREATE VIEW Customer_Loan_View AS SELECT C.Customer_ID, C.Customer_Name, L.Loan_ID, L....	0 row(s) affected
5	20:18:36	SELECT * FROM Customer_Loan_View LIMIT 0, 1000	0 row(s) returned
6	20:19:14	CREATE VIEW Transaction_View AS SELECT T.Transaction_ID, T.Account_No, C.Customer_Name, ...	0 row(s) affected
7	20:19:14	SELECT * FROM Transaction_View LIMIT 0, 1000	0 row(s) returned
8	20:21:34	CREATE TRIGGER CheckBalance BEFORE UPDATE ON Account FOR EACH ROW BEGIN IF NEW.Balan...	0 row(s) affected

Object Info Session

Trigger 2 — Prevent Invalid Transaction Amount

MySQL Workbench

Rahul_2520090016 x

File Edit View Query Database Server Tools Scripting Help

Navigator skill week - 3.4* x

Limit to 1000 rows

270 SIGNAL SQLSTATE '45000'

271 SET MESSAGE_TEXT = 'Transaction failed: Insufficient balance';

272 END IF;

273 END //

274

275 DELIMITER ;

276

277 DELIMITER //

278

279 CREATE TRIGGER CheckTransactionAmount

280 BEFORE INSERT ON Bank_Transaction

281 FOR EACH ROW

282 BEGIN

283 IF NEW.Amount <= 0 THEN

284 SIGNAL SQLSTATE '45000'

285 SET MESSAGE_TEXT = 'Transaction amount must be greater than zero';

286 END IF;

287 END //

288

289 DELIMITER ;

Administration Schemas

Information:

Schema: bookflow_db

Object Info Session

Output

Action Output

#	Time	Action	Message
2	20:17:55	CREATE VIEW Customer_Account_View AS SELECT C.Customer_ID, C.Customer_Name, C.Phone, ...	0 row(s) affected
3	20:17:55	SELECT * FROM Customer_Account_View LIMIT 0, 1000	0 row(s) returned
4	20:18:36	CREATE VIEW Customer_Loan_View AS SELECT C.Customer_ID, C.Customer_Name, L.Loan_ID, ...	0 row(s) affected
5	20:18:36	SELECT * FROM Customer_Loan_View LIMIT 0, 1000	0 row(s) returned
6	20:19:14	CREATE VIEW Transaction_View AS SELECT T.Transaction_ID, T.Account_No, C.Customer_Name, ...	0 row(s) affected
7	20:19:14	SEL SELECT * FROM Transaction_View	0 row(s) returned
8	20:21:34	CR LIMIT 0, 1000	0 row(s) affected
9	20:22:34	CREATE TRIGGER CheckTransactionAmount BEFORE INSERT ON Bank_Transaction FOR EACH ROW B...	0 row(s) affected

Trigger 3 — Transaction Audit

MySQL Workbench

Rahul_2520090016 x

File Edit View Query Database Server Tools Scripting Help

Navigator skill week - 3.4 x

Limit to 1000 rows

SCHEMAS

Filter objects

- 2520090016_rahul
 - bankdb**
 - Tables
 - Views
 - Stored Procedures
 - Functions
 - bookflow_db
 - joins_set_operations
 - klu
 - rahul_2520090016
 - sys
 - workflow_db

Administration Schemas

Information

Schema: bookflow_db

```
272     END IF;
273   END //
274
275   DELIMITER ;
276
277   DELIMITER //
278
279   CREATE TRIGGER CheckTransactionAmount
280   BEFORE INSERT ON Bank_Transaction
281   FOR EACH ROW
282   BEGIN
283     IF NEW.Amount <= 0 THEN
284       SIGNAL SQLSTATE '45000'
285       SET MESSAGE_TEXT = 'Transaction amount must be greater than zero';
286     END IF;
287   END //
288
289   DELIMITER ;
290
291   STX
```

Output

Action Output

#	Time	Action	Message
3	20:17:55	SELECT * FROM Customer_Account_View LIMIT 0, 1000	0 row(s) returned
4	20:18:36	CREATE VIEW Customer_Loan_View AS SELECT C.Customer_ID, C.Customer_Name, L.Loan_ID, ...	0 row(s) affected
5	20:18:36	SELECT * FROM Customer_Loan_View LIMIT 0, 1000	0 row(s) returned
6	20:19:14	CREATE VIEW Transaction_View AS SELECT T.Transaction_ID, T.Account_No, C.Customer_Name, ...	0 row(s) affected
7	20:19:14	SELECT * FROM Transaction_View LIMIT 0, 1000	0 row(s) returned
8	20:21:34	CREATE TRIGGER CheckBalance BEFORE UPDATE ON Account FOR EACH ROW BEGIN IF NEW.Bal...	0 row(s) affected
9	20:22:34	CREATE TRIGGER CheckTransactionAmount BEFORE INSERT ON Bank_Transaction FOR EACH ROW B...	0 row(s) affected
10	20:23:16	CREATE TABLE Transaction_Audit (Audit_ID INT PRIMARY KEY AUTO_INCREMENT, Transaction_ID...	0 row(s) affected

Object Info Session

MySQL Workbench

Rahul_2520090016 x

File Edit View Query Database Server Tools Scripting Help

Navigator: skill week - 3.4* x

Limit to 1000 rows

290 DELIMITER //

291

292 CREATE TRIGGER TransactionAudit

293 AFTER INSERT ON Bank_Transaction

294 FOR EACH ROW

295 BEGIN

296 INSERT INTO Transaction_Audit

297 (

298 Transaction_ID,

299 Account_No,

300 Transaction_Type,

301 Amount

302)

303 VALUES

304 (

305 NEW.Transaction_ID,

306 NEW.Account_No,

307 NEW.Transaction_Type,

308 NEW.Amount

309);

Administration Schemas

Information: Schema: bookflow_db

Object Info Session

Output:

Action Output

#	Time	Action	Message
4	20:18:36	CREATE VIEW Customer_Loan_View AS SELECT C.Customer_ID, C.Customer_Name, L.Loan_ID, ...	0 row(s) affected
5	20:18:36	SELECT * FROM Customer_Loan_View LIMIT 0, 1000	0 row(s) returned
6	20:19:14	CREATE VIEW Transaction_View AS SELECT T.Transaction_ID, T.Account_No, C.Customer_Name, ...	0 row(s) affected
7	20:19:14	SELECT * FROM Transaction_View LIMIT 0, 1000	0 row(s) returned
8	20:21:34	CREATE TRIGGER CheckBalance BEFORE UPDATE ON Account FOR EACH ROW BEGIN IF NEW.Bal...	0 row(s) affected
9	20:22:34	CREATE TRIGGER CheckTransactionAmount BEFORE INSERT ON Bank_Transaction FOR EACH ROW B...	0 row(s) affected
10	20:23:16	CREATE TABLE Transaction_Audit (Audit_ID INT PRIMARY KEY AUTO_INCREMENT, Transaction_ID...	0 row(s) affected
11	20:25:11	CREATE TRIGGER TransactionAudit AFTER INSERT ON Bank_Transaction FOR EACH ROW BEGIN IN...	0 row(s) affected

PROCEDURES

MySQL Workbench interface showing a SQL script execution session.

Navigator:

- 2520090016_rahul
 - bankdb
 - Tables
 - Views
 - Stored Procedures
 - Functions
 - bookflow_db
 - joins_set_operations
 - klu
 - rahul_2520090016
 - sys
 - workflow_db

SQL Script (skill week - 3.4*):

```
451 • CALL DepositMoney(10001, 5000);
452
453 • CALL WithdrawMoney(10001, 3000);
454
455 • CALL TransferMoney(10001, 10002, 5000);
456
457 • CALL GetCustomerLoans(101);
458
459 • CALL HighBalanceAccounts(50000);
460
461 • CALL GetBalance(10001, @CurrentBalance);
462
463 • SELECT @CurrentBalance;
```

Result Grid:

Result	Grid	Filter Rows:	Export:	Wrap Cell Content:
@CurrentBalance				
NULL				

Schema: bookflow_db

Output:

Action Output

#	Time	Action	Message
✓ 33	20:28:02	CALL GetCustomerAccounts(101)	0 row(s) returned
✓ 34	20:28:02	CALL DepositMoney(10001, 5000)	0 row(s) affected
✓ 35	20:28:02	CALL WithdrawMoney(10001, 3000)	0 row(s) affected
✓ 36	20:28:02	CALL TransferMoney(10001, 10002, 5000)	0 row(s) affected
✓ 37	20:28:02	CALL GetCustomerLoans(101)	0 row(s) returned
✓ 38	20:28:02	CALL HighBalanceAccounts(50000)	0 row(s) returned
⚠ 39	20:28:02	CALL GetBalance(10001, @CurrentBalance)	0 row(s) affected, 1 warning(s):
✓ 40	20:28:02	SELECT @CurrentBalance LIMIT 0, 1000	1 row(s) returned

Object Info: Session